

A rank-one counterexample to conjecture 00000001952

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The original assertion

The conjecture defines the minimum index of a finite-index subgroup of $\text{Out}(F_n)$ and claims its value is $2^n \binom{n}{2}$. It gives no restriction $n \geq 2$ or $n \geq 3$. We use the positive rank $n = 1$, for which the asserted value is $2 \binom{1}{2} = 0$.

The group is not replaced by a numerical proxy. Set F_n equal to the free group on n generators, let $\text{Inn}(F_n)$ be the image of the conjugation homomorphism $F_n \rightarrow \text{Aut}(F_n)$, and define $\text{Out}(F_n) = \text{Aut}(F_n)/\text{Inn}(F_n)$. For any automorphism α , conjugating the inner automorphism $c_g : x \mapsto gxg^{-1}$ gives $\alpha c_g \alpha^{-1} = c_{\alpha(g)}$, proving that the image is normal and that this is the actual outer automorphism group.

Theorem 1. *The proposed rank-one minimum-index value is false, both if all finite-index subgroups are permitted and if only proper ones are permitted.*

Proof. For any group G and subgroup H , the left-coset space G/H is nonempty: it contains the identity coset. If it is finite, its cardinality $[G : H]$ is therefore a positive integer. If all finite-index subgroups are permitted, G itself has index 1. Consequently the minimum finite subgroup index is exactly 1, including for the actual group $G = \text{Out}(F_1)$, whereas the displayed formula is 0.

If “subgroup” is instead understood to mean “proper subgroup”, any attained finite minimum must still be positive because its witnessing subgroup has a nonempty finite coset space. It cannot be 0 either. This second argument does not assert a particular proper minimum, or assume a theorem identifying $\text{Out}(F_1)$ with another group. Thus neither reading can give the conjectured value. $\square$

The extra claim about smaller indices arising as kernels cannot repair a false numerical minimum. The counterexample is to the assertion as written; it makes no claim about a modified problem restricted to higher rank.

Formal verification and prior submission

Lean uses Mathlib’s `FreeGroup` (`Fin n`), `MulAut`, actual conjugation homomorphism, its image subgroup, normality, quotient group, and subgroup index. It proves the equivalence of finite quotient and nonzero index. Both minimum predicates require an actual witnessing subgroup attaining the minimum, as well as minimality. The capstones `conjecture_01952_false_at_one` and its `proper` version negate the formula in these two conventions on the actual group.

Previous closed PR #240 used a rank-three determinant-kernel argument, but its Lean proof was only a numerical comparison and contained none of the relevant groups, subgroups, or indices. The reviewer also noted that the problem already mentions action kernels. This submission does not reuse that determinant argument or its numeric certificate: it formally constructs the actual outer automorphism group and refutes the rank-one formula via genuine finite coset indices. The project compiles on Lean 4.33.1 with pinned Mathlib, no unfinished proofs, no native evaluation, and no additional axioms. The only printed foundational dependencies are propositional extensionality, classical choice, and quotient soundness. No brute-force computation is needed.